

100 Days of Azure DevOps

Day 46 Handout — init / plan / apply / destroy. State is the mapping from resource addresses to Azure IDs — not a cache

5 - Infrastructure as Code · Personal lab only · Educational content · Not a sales pitch

Architecture A — four verbs and the file that remembers

Verb	What it does	Touches Azure?
terraform init	Download providers; later: configure backend	No (unless backend init migrates state)
terraform plan	Diff: desired config vs state vs reality (refresh)	Read. Does not mutate (except refresh)
terraform apply	Execute the plan; write terraform.tfstate	Yes
terraform destroy	Plan a teardown; remove from state as it deletes	Yes. Adult ending for a lab
terraform.tfstate	Address (azure_terraform_resource_group.lab) → ARM resource ID	Lose it = ghosts: create already-exists, destroy skips live RGs

Architecture B — local state is a lab, not a team

Setup	OK when	Move off it when
Local tfstate	Today: one RG, one human, destroy tonight	Two laptops / two applies (Day 48 remote + lock)
Bicep primary	Still run the four verbs once so state is not a rumor	Skipping Terraform entirely after Day 41 ADR
-auto-approve	destroy on a disposable RG	apply to anything you would miss
Partial apply	Never as a habit	You now have state that does not match the file

Architecture C — pitfalls that schedule amnesia

Pitfall	What you see	Fix
Deleted tfstate to 'clean up'	Next apply wants to create the RG that still exists	Never delete state to tidy. destroy, or backend migrate
apply without plan	Mental diff missed a replace	Always read plan. Surprise = no apply
Skipped destroy	Empty RG becomes a subscription hobby	terraform destroy -auto-approve tonight
Committed tfstate with secrets	Attributes in git history	gitignore terraform.tfstate*; remote backend later

One-liner to remember

Terraform state is the memory of your infra — lose it and you are arguing with ghosts.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal PC + personal subscription. Local state on purpose. Always destroy at end of night. Do not commit tfstate.

- 1 Install Terraform (terraform -version). Confirm it is on PATH.
- 2 New folder infra/tf-day46/. gitignore terraform.tfstate and .terraform/. Write a resource group only (starter).
- 3 terraform init then terraform plan. Read the plan. Then terraform apply (type yes or -auto-approve on this disposable RG).
- 4 az group show -n rg-day46-tf. Confirm reality matches state (terraform state list).
- 5 Do not delete terraform.tfstate in Explorer. That is how ghosts start.
- 6 terraform destroy -auto-approve. Confirm the RG is gone. If destroy fails, do not shrug — fix it tonight.

Local-state lab — RG only, then destroy

```
# infra/tf-day46/main.tf
terraform {
  required_version = ">= 1.6.0"
  required_providers {
    azurerm = {
      source  = "hashicorp/azurerm"
      version = "~> 3.100"
    }
  }
}
provider "azurerm" { features {} }
resource "azurerm_resource_group" "lab" {
  name     = "rg-day46-tf"
  location = "Central India"
}

# az login    # personal subscription
# terraform init
# terraform plan
# terraform apply
# terraform destroy -auto-approve
```

Done checklist

- init → plan → apply → destroy all ran
- tfstate is gitignored, not deleted as cleanup
- RG gone at the end of the night
- Can explain state as ID mapping, not a cache

Tomorrow — Day 47

Terraform with Azure (azurerm) — features {}, CLI auth, RG + storage. No subscription-Owner robot.